Sahaj Bhadja

(213) 284-4651 sahaj.bhadja@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Product-minded Software Engineer with 2+ years owning **React/TypeScript interfaces, Python services, PostgreSQL data models, and AI-enabled workflows** from ambiguous user problems through design, implementation, release, feedback, and production support.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Advanced NLP, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Structures, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Software Engineer *Los Angeles, CA*

- Owned an AI education platform for 500+ users from stakeholder discovery through release, reducing dashboard latency 71% through **workflow design, normalized PostgreSQL-compatible schemas, indexed queries, and iterative product feedback**.
- Built asynchronous FastAPI orchestration processing 1K+ concurrent jobs at 3x throughput through **stateful workflows, failure isolation, background processing, monitoring, production debugging, and independent end-to-end complex technical decision-making**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Designed React and Flask workflows for uploads, case history, model controls, and feedback, supporting 150+ instructors and 10K+ daily requests through **rapid prototypes, usability iteration, and production releases**.
- Built concurrent Python ingestion for PDF, DOCX, ZIP, OCR, and multimodal routing, reducing turnaround 60% while coordinating **frontend, backend, data validation, testing, deployment, and user-facing error recovery**.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Developed Decision Transformer pipelines processing 100K+ autonomous-driving trajectories, improving planning 18% through **Python experimentation, dense temporal data modeling, reproducible evaluation, and quantitative interactive visualization of model behavior**.
- Integrated HD-map geometry, lane context, and ego trajectories with multi-head attention, improving closed-loop stability 15% while **translating scientific questions into interpretable interfaces, metrics, and documented research findings**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Designed cross-platform APIs and cloud data pipelines using object-oriented patterns, AWS Glue, S3, and SQL, improving response times 30% while reducing technical debt 20% through **product-oriented service boundaries**.
- Automated 20+ unit and integration suites across Windows and Linux using PyTest, Mockito, Jenkins, and virtual machines, strengthening **release confidence, regression coverage, cross-functional reviews, and maintainable product delivery**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built React and Node.js workflows for resources, billing, permissions, and notifications, synchronizing 2M+ records through **reusable TypeScript components, REST APIs, WebSockets, authentication, and a shared operational database**.
- Automated Docker and Jenkins deployments, reducing release time 50% while using Sentry, Azure Monitor, and Kibana for **user-impact investigation, product analytics, root-cause debugging, and reliable weekly releases**.

Skills

Product Engineering: Product Discovery, User Interviews, Workflow Design, Rapid Prototyping, UX Design, Visual Design, Design Systems, Product Analytics, User Feedback, End-to-End Ownership
Frontend: React, TypeScript, JavaScript, Next.js, HTML5, CSS3, Responsive Design, Accessibility, Data Visualization, Component Architecture, State Management
Backend & Data: Python, FastAPI, Flask, Node.js, PostgreSQL, SQL, REST APIs, WebSockets, Data Modeling, Background Jobs, Authentication, Role-Based Access
Scientific & AI: PyTorch, Transformers, LLMs, RAG, Machine Learning, Experimental Design, Scientific Dashboards, High-Dimensional Data, Model Evaluation, Explainability
Delivery: AWS, Azure, GCP, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Git, Unit Testing, Integration Testing, Monitoring, Debugging, Agile, Technical Documentation

Projects

Personalized Vaccine Operations Workspace | *React, TypeScript, Python, PostgreSQL*

- Designed a **patient- and clinician-facing product** connecting case intake, sequencing status, antigen review, vaccine design, manufacturing handoffs, follow-ups, and role-specific timelines through one auditable shared data model.

Neoantigen Design Studio | *React, TypeScript, FastAPI, PostgreSQL, Plotly*

- Built a **dense scientific decision interface** for ranking antigen candidates, comparing prediction evidence, filtering assay results, annotating design choices, and exporting review-ready vaccine constructs with provenance and collaborative approvals.

Big-Brain | *LLaMA 2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs* | Live

- Built a **retrieval-augmented AI knowledge product** indexing 50K+ documents with LLaMA 2, FAISS, Pinecone, PostgreSQL, and REST APIs to deliver grounded answers through a deployed GCP-backed user experience.